

C interview questions and answers:

1. What is C language?

- C language is a structure/procedure oriented, middle level programming language developed at Bell Laboratories in 1972 by Dennis Ritchie.
- C language was invented for implementing UNIX operating system.
- In 1978, Dennis Ritchie and Brian Kernighan published the first edition "The C Programming Language".
- Also, C language is an ANSI/ISO standard and powerful programming language for developing real time applications

2. Who developed C language?

C programming language was developed at Bell Laboratories in 1972 by Dennis Ritchie.

3. Describe about history of C programming language.

- C programming language is a structure oriented programming language, was developed at Bell Laboratories in 1972 by Dennis Ritchie
- C language features were derived from earlier language called "B" (Basic Combined Programming Language – BCPL)
- In earlier days, programs were written in assembly level language. So, it had happened to write very big programs to perform specific tasks using assembly code.
- But, 'B' language could perform the same task in few lines of program and it was faster than assemble language code.
- But, B language did not support some features like data types and structures etc. So, this was a drawback of B language. So, Dennis Ritchie developed C language by keeping most part of the B language and adding many features that produced powerful and effective outputs.
- So, C language was invented for implementing UNIX operating system. Most of the UNIX components were rewritten in C.
- In 1978, Dennis Ritchie and Brian Kernighan published the first edition "The C Programming Language" and commonly known as K&R C
- In 1983, the American National Standards Institute (ANSI) established a committee to provide a modern, comprehensive definition of C. The resulting definition, the ANSI standard, or "ANSI C", was completed late 1988.
- C89/C90 standard – First standardized specification for C language was developed by American National Standards Institute in 1989. C89 and C90 standards refer to the same programming language.
- C99 standard – Next revision was published in 1999 that introduced new futures like advanced data types and other changes.
- C11 standard adds new features to C and library like type generic macros, anonymous structures, improved Unicode support, atomic operations, multi-threading, and bounds-checked functions. It also makes some portions of the existing C99 library optional, and improves compatibility with C++.
- Embedded C includes features not available in normal C like fixed-point arithmetic, named address spaces, and basic I/O hardware addressing.

4. Where is C programming language used or uses of C language?

C language is used to develop system applications that forms major portion of operating systems such as Windows, UNIX and Linux. Operating systems, C compiler and all UNIX application programs are written in C language. Below are some examples of uses of C language.

- Database systems
- Graphics packages
- Word processors
- Spread sheets
- Operating system development
- Compilers and Assemblers
- Network drivers
- Interpreters

5. What is the difference between C and C++?

Even though C and C++ programming languages are belonging to middle level languages, both are differed in below.

- C is structure/procedure oriented programming language whereas C++ is object oriented programming language.
- C language program design is top down approach whereas C++ is using bottom up approach.
- Polymorphism, virtual function, inheritance, Operator overloading, namespace concepts are not available in C programming language. Whereas C++ language supports all these concepts and features.
- C language gives importance to functions rather than data. Whereas C++ gives importance to data rather than functions.
- So, data and function mapping is difficult in C. But, data and function mapping is simple in C++ that can be done using objects.
- C language does not support user define data types. Whereas C++ supports user define data types.
- Exception handling is not present in C programming language. Whereas exception handling is present in C++ language.
- C language allows data to freely flow around the functions. But, data and functions are bound together in C++ which does not allow data to freely flow around the functions.

6. What is the difference between top down approach and bottom up approach in programming languages?

Top down approach and bottom up approach are involved in software development. These approaches are not involved in program execution.

- Structure/procedure oriented programming languages like C programming language follows top down approach. Whereas object oriented programming languages like C++ and Java programming language follows bottom up approach.
- Top down approach begins with high level design and ends with low level design or development. Whereas, bottom up approach begins with low level design or development and ends with high level design.
- In top down approach, main() function is written first and all sub functions are called from main function. Then, sub functions are written based on the requirement. Whereas, in bottom

up approach, code is developed for modules and then these modules are integrated with main() function.

- Now-a-days, both approaches are combined together and followed in modern software design.

7. What is the difference between C and Java?

- C is structure/procedure oriented programming language whereas Java is object oriented programming language.
- C language program design is top down approach whereas Java is using bottom up approach.
- C language is middle level language whereas Java is high level language.
- Exception handling is not present in C programming language. Whereas exception handling is present in Java.
- Polymorphism, virtual function, inheritance, Operator overloading, namespace concepts are not available in C programming language. Whereas Java supports all these concepts and features.

8. C language has been developed in which language?

C language has been developed using assembly level language.

9. Which year C language is developed?

C programming language was developed at Bell Laboratories in 1972 by Dennis Ritchie.

10. What is meant by programming language and give some examples?

- Programming language is nothing but a language designed to communicate to machines through instructions and commands.
- Normally machines are computers. Programs are written using some programming languages to control the behaviour of machines/computers and to make them to perform required tasks.
- Programming language example: Assembly language, C language, C++ language, Java, C#, .NET, Python etc.

11. Describe about C standards.

- C89/C90 standard – First standardized specification for C language was developed by American National Standards Institute in 1989. C89 and C90 standards refer to the same programming language.
- C99 standard – Next revision was published in 1999 that introduced new features like advanced data types and other changes.
- C11 standard adds new features to C and library like type generic macros, anonymous structures, improved Unicode support, atomic operations, multi-threading, and bounds-checked functions. It also makes some portions of the existing C99 library optional, and improves compatibility with C++.

12. What are the key features or characteristics of C language?

- Reliability

- Portability
- Flexibility
- Interactivity
- Modularity
- Efficiency and Effectiveness

13. What is embedded C?

- Embedded C is the extension of C programming language.
- Embedded C is used to develop micro controller based applications.
- Embedded C includes features not available in normal C like fixed-point arithmetic, named address spaces, and basic I/O hardware addressing.
- Cell phones, MP3 players are some example for embedded systems in which embedded C is used to program and control these devices.

14. Which level is C language belonging to?

- C language is belonging to middle level language. C language behaves as a bridge between machine level (low level) languages and high level languages.
- C language is more user friendly than machine level languages. And, C language does not support all the concepts that high level languages offer. So, C programming language is called as middle level language.

15. What do you mean by high level, middle level and low level languages and give an example for each?

High level languages –

- These level languages provide almost everything that the programmer might need to do as already build into the language.
- Example: Java, Python

Middle level languages –

- These languages don't provide all the built-in functions found in high level languages, but provide all building blocks that we need to produce the result we want.
- Example: C, C++

Low level languages –

- These languages provide nothing other than access to the machine's basic instruction set.
- Example: Assembly language

16. What is the difference between structured oriented, object oriented and non-structure oriented programming language?

Structured oriented programming language –

- In this type of language, large programs are divided into small programs called functions.
- Prime focus is on functions and procedures that operate on data
- Data moves freely around the systems from one function to another

- Program structure follows “Top Down Approach”
- Example: C, Pascal, ALGOL and Modula-2

Object oriented programming language –

- In this type of language, programs are divided into objects
- Prime focus is on the data that is being operated and not on the functions or procedures
- Data is hidden and cannot be accessed by external functions
- Program structure follows “Bottom UP Approach”
- Example: C++, JAVA and C# (C sharp)

Non-structure oriented programming language –

- There is no specific structure for programming this language.
- Example: BASIC, COBOL, FORTRAN

17. What is compiler?

- Compiler is a program that converts human readable code into machine readable code. This process is called compilation.

18. What is the difference between assembler, compiler and interpreter?

- Assembler is a program that converts assembly level language (low level language) into machine level language.
- Compiler compiles entire C source code into machine code. Whereas, interpreters converts source code into intermediate code and then this intermediate code is executed line by line.

19. What is printf()?

- printf() is an inbuilt library function in C which is available in C library by default. This function is declared and related macros are defined in “stdio.h” header file.
- printf() function is used to print the “character, string, float, integer, octal and hexadecimal values” onto the output screen.

20. What is scanf()?

- scanf() function is an inbuilt library function in C which is available in C library by default. This function is declared and related macros are defined in “stdio.h” header file.
- scanf() function is used to read character, string, numeric data from keyboard.

21. What is meant by protocol?

- Protocol is nothing but a set of rules to be followed by a programmer.

22. Execution of a C program starts from which function?

Always, execution of a C program starts from main() function.

23. What are all the sections that a C program may/must have?

There are many sections in a C program structure. They are,

- Documentation section
- Link Section
- Definition Section
- Global declaration section
- Function prototype declaration section
- Main function
- User defined function section

main() function section is the important section in a C program as program execution starts from main() function only in C language. A C program may not have all other sections except main() function.

24. What is IDE?

- IDE is nothing but Integrated Development Environment. IDE is a tool that provides user interface with compilers to create, compile and execute C programs.
- Example: Turbo C++, Borland C++ and DevC++. These provide Integrated Development Environment with compiler for both C and C++ programming language.

25. List out some of C compilers.

There are so many compilers available in market for Windows operating system and UNIX. We are listing some of them here for your reference.

- AMPC
- CCS C Compiler
- ch
- clang
- Cygwin
- Digital mars
- GCC compiler
- MikroC Compiler
- Portable C Compiler, Power C, QuickC, Ritchie C Compiler, Small-C

26. What is header file in C language?

- Header file is a file that contains function declaration and macro definition for C in-built library functions.
- All C standard library functions are declared in many header files which are saved as file_name.h.
- We are including these header files in our C program using “#include <file_name.h>” command to make use of the functions those are declared in the header files.
- When we include header files in our C program using “#include <filename.h>” command, all C code of the header files are included in C program. Then, this C program is compiled by compiler and executed.

27. Is C language case sensitive?

Yes. C language instructions/commands/functions and everything used in C program are case sensitive.

28. What is Macro? Why do we use macro?

- Macro is a name which is given to a value or to a piece of code/block in a program. Instead of using the value, we can use macro which will replace the value in a program.
- The reason for using macro is, consider below example.
- You are using a person's age as 50 in many places in your program. If you use direct value in all the places in your program, it is very difficult to change the age value in future if you want to change it to 60. If you use macro in your program, it is very simple to replace the value and we can change in only one place which will change the value in all places in your program.

Syntax:	#define		<MACRO_NAME>		VALUE
Example:					
Original	declaration	–	#define	AGE	50
Modified declaration – #define AGE 60					

29. What is data type in C?

- Data types in C language are defined as the data storage format that a variable can store a data to perform a specific operation.
- Data types are used to define a variable before to use in a program.
- Size of variable, constant and array are determined by data types.

30. What is the difference between int, char, float & double data types?

- Integer data type allows a variable to store numeric values. The storage size of int data type is 2 or 4 or 8 byte. It varies depend upon the processor in the CPU.
- Character data type allows a variable to store only one character. Storage size of character data type is 1.
- Float data type allows a variable to store decimal values. Storage size of float data type is 4. This also varies depend upon the processor in the CP.
- Double data type is also same as float data type which allows up-to 10 digits after decimal.

31. What is the use of sizeof() function in C?

sizeof() function is used to find the memory space allocated for each data type in C.

32. What is modifier in C?

- The amount of memory space to be allocated for a variable is derived by modifiers.
- Modifiers are prefixed with basic data types to modify (either increase or decrease) the amount of storage space allocated to a variable.
- For example, storage space for int data type is 4 byte for 32 bit processor. We can increase the range by using long int which is 8 byte. We can decrease the range by using short int which is 2 byte.

33. What are different types of modifiers in C?

There are 5 modifiers available in C language. They are,

- short
- long
- signed
- unsigned
- long long

34. What is enum in C?

- Enumeration is a data type that consists of named integer constants as a list.
- It start with 0 (zero) by default and value is incremented by 1 for the sequential identifiers in the list.

35. What is void in C?

Void is an empty data type that has no value.

- We use void data type in functions when we don't want to return any value to the calling function.

Example:

void sum (int a, int b); – This function won't return any value to the calling function.

int sum (int a, int b); – This function will return value to the calling function.

- We use void data type in pointer like "void *p". It means, pointer "p" is neither pointing to int data type nor char data type. It acts as generic pointer. We are using void pointer when we are not sure on the data type that this pointer will point to. We can use void pointer to refer either integer data or char data. But, this void pointer should not be dereferenced without explicit type casting.
- We use void in functions as "int function_name (void)". Here void means, this function does not pass any argument.

36. What is token in C?

- C tokens are the basic buildings blocks in C language which are constructed together to write a C program.
- Each and every smallest individual unit in a C program are known as C tokens.

37. What are the types of C tokens?

C tokens are of six types. They are,

1. Keywords (eg: int, while),
2. Identifiers (eg: main, total),
3. Constants (eg: 10, 20),
4. Strings (eg: "total", "hello"),
5. Special symbols (eg: {}, {}),
6. Operators (eg: +, /, -, *)

38. What is identifier in C?

- Each program elements in a C program are given a name called identifiers.
- Names given to identify Variables, functions and arrays are examples for identifiers.
- Example: x is a name given to integer variable in above program

39. What is keyword in C?

- Keywords are pre-defined words in a C compiler. Each keyword is meant to perform a specific function in a C program.
- Since keywords are referred names for compiler, they can't be used as variable name.

40. List out some keywords available in C language.

- Below are some of keywords that C language offers.
- auto, double, int, struct, break, else, long, switch, case, enum, register, typedef, char, extern, return, union, const, float, short, unsigned, continue, for, signed, void, default, goto, sizeof, volatile, do, , if, static, while

41. What is constant in C?

- Constants refer to fixed values. They are also called as literals.
- C Constants are also like normal variables. But, only difference is, constant values can't be modified by the program once they are defined. Constants may be belonging to any of the data type.

42. What are the types of constants in C?

- Integer constants
- Real or Floating point constants
- Octal & Hexadecimal constants
- Character constants
- String constants
- Backslash character constants

43. What is variable in C?

- C variable is a named location in a memory where a program can manipulate the data. This location is used to hold the value of the variable.
- The value of the C variable may get change in the program.
- C variable might be belonging to any of the data type like int, float, char etc.

44. What is the difference between constant and variable in C?

Constant values can't be modified by the program once it is declared. But, variable values can be modified by the program.

45. Can variable name start with numbers?

No. Variable names can't start with numbers as per the variable naming rule.

46. What is the difference between variable declaration and variable definition in C?

- Variable declaration tells the compiler about data type and size of the variable. Whereas, variable definition allocates memory to the variable
- Variable can be declared many times in a program. But, definition can happen only one time for a variable in a program.
- Variable declaration is for assignment of properties and identification to a variable. Whereas, variable definition is for assignments of storage space to a variable

47. What are the different types of variable in C?

1. Local variable
2. Global variable
3. Environment variable

48. What is local variable in C?

- The variables which are having scope/life only within the function are called local variables.
- These variables are declared within the function and can't be accessed outside the function.

49. What is global variable in C?

- The variables which are having scope/life throughout the program are called global variables.
- Global variable is defined outside the main function. So, this variable is visible to main function and all other sub functions.

50. What is environment variable in C?

- Environment variable is a variable that will be available for all C applications and C programs.
- Once environment variables are exported, we can access them from anywhere in a C program without declaring and initializing in an application or C program.

51. What is operator in C?

- The symbols which are used to perform logical and mathematical operations in a C program are called C operators.
- These C operators join individual constants and variables to form expressions. Operators, functions, constants and variables are combined together to form expressions.

52. What are the different types of operators in C?

C language offers many types of operators. They are,

1. Arithmetic operators
2. Assignment operators
3. Relational operators
4. Logical operators
5. Bit wise operators
6. Conditional operators (ternary operators)
7. Increment/decrement operators

8. Special operators

53. What is the syntax for ternary operator in C?

- Ternary operator is same as if else control statement in C.
- Syntax : (Condition? true_value: false_value);
- Example: (A > 100? 0: 1);

54. What is arithmetic operator in C?

- C Arithmetic operators are used to perform mathematical calculations like addition, subtraction, multiplication, division and modulus in C programs.
- Arithmetic operators are +, -, *, /, %

55. What is assignment operator in C?

- Assignment operators are used to assign the values to the variables.
- Assignment operators are =, +=, -=, /=, %= etc.

56. What is the relational operator in C?

- Relational operators are used to find the relation between two variables. i.e. to compare the values of two variables in a C program.
- Relational operators are >, <, >=, <=, ==, !=

57. What is the logical operator in C?

- Logical operators are used to perform logical operations on the given expressions.
- There are 3 logical operators in C language. They are,
 1. logical AND (&&)
 2. logical OR (||)
 3. logical NOT (!)

58. What is bitwise operator in C?

- Bitwise operators are used to perform bit operations. Decimal values are converted into binary values which are the sequence of bits and bit wise operators work on these bits.
- Bit wise operators in C language are & (bitwise AND), | (bitwise OR), ~ (bitwise NOT), ^ (XOR), << (left shift) and >> (right shift).

59. What are all decision control statements in C?

There are 3 types of decision making control statements in C language. They are,

1. if statements
2. if else statements
3. nested if statements

60. What are all loop control statements in C?

- Loop control statements in C are used to perform looping operations until the given condition is true. Control comes out of the loop statements once condition becomes false.
- There are 3 types of loop control statements in C language. They are,

1. for
2. while
3. do-while

61. What is the difference between while and do-while loops in C?

- While loop is executed only when given condition is true.
- Whereas, do-while loop is executed for first time irrespective of the condition. After executing while loop for first time, then condition is checked.

62. What is the difference between single equal “=” and double equal “==” operators in C?

- Single equal is an assignment operator used to assign the values to the variables.
- But, double equal is relational operator used to compare two variable values whether they are equal or not.

63. What is the difference between pre increment operator and post increment operator in C?

- Pre increment operator is used to increment variable value by 1 before assigning the value to the variable.
- Post increment operator is used to increment variable value by 1 after assigning the value to the variable.

64. What is the difference between pre decrement operator and post decrement operator?

- Pre decrement operator is used to decrement variable value by 1 before assigning the value to the variable.
- Post decrement operator is used to decrement variable value by 1 after assigning the value to the variable.

65. What is “&” and “*” operators in C?

- “*” Operator is used as pointer to a variable. Example: *a where * is pointer to the variable a.
- & operator is used to get the address of the variable. Example: &a will give address of a.

66. What will happen if break statement is not used in switch case in C?

- Switch case statements are used to execute only specific case statements based on the switch expression.
- If we do not use break statement at the end of each case, program will execute all consecutive case statements until it finds next break statement or till the end of switch case block.

67. Why is default statement used in switch case in C?

Switch case statements are used to execute only specific case statements based on the switch expression. If switch expression does not match with any case, default statements are executed by the program.

68. What is the use of “goto” statement?

goto statement is used to transfer the normal flow of a program to the specified label in the program. Below is the syntax for goto statement in C.

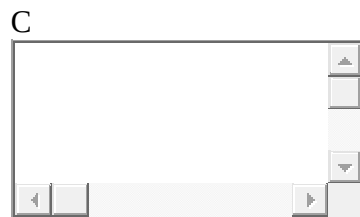
```
{
.....
goto label;
.....
.....
LABEL:
statements;
}
```

69. What value will be assigned to the variable X if a = 10, b = 20, c = 30, d = 40 in below expression?

$X = a/b + c*d - c;$

- The above arithmetic operation is performed based on the precedence of the operators.
- In above mentioned expression, $c*d$ will be performed first. Then, a/b , then $(c*d) - c$, then $(a/b) + ((c*d) - c)$.
- Please check the operator precedence table to know the priority and associativity of the C operators.
- Output of the above expression is 1170.

Example program:



```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 int main()
5 {
6     int a = 10, b = 20, c = 30, d = 40, X ;
7
8     X = a/b+c*d-c;
9     printf(" Value of X = %d\n", X);
10    return 0;
11 }
```

Output : Value of X = 1170

70. What value is assigned to the below variables?

```
int X1 = 13/3;  
int X2 = 13%3;
```

- The value of X1 is 4
- The value of X2 is 1

71. What is the difference between auto variable and register variable in C?

Storage class of all variables are auto by default unless we specify a variable is register or static or extern in C program.

- Both auto variable and register variable are local variables. Register variables are stored in register memory. Whereas, auto variables are stored in main CPU memory.
- Register variables will be accessed very faster than the normal/auto variables since they are stored in register memory rather than main memory.
- But, only limited variables can be used as register since register size is very low. (16 bits, 32 bits or 64 bits)

72. What is the difference between auto variable and static variable in C?

- Both auto and static variables are local variables.
- Static variables can retain the value of the variable between different function calls.
- But, scope of auto variable is within the function only. It can't retain the value of the variable between different function calls.

73. Where should type cast function not be used in C?

- Type cast function should not be used in const and volatile declaration.
- Because, constant value can't be modified by the program once they are defined.
- And, volatile variable values might keep on changing without any explicit assignment by the program as operating system will be modifying these values.

74. How many arguments can be passed to a function in C?

- Any number of arguments can be passed to a function. There is no limit on this. Function arguments are stored in stack memory rather than heap memory. Stack memory allocation is depending on the operating system.
- So, any number of arguments can be passed to a function as much as stack has enough memory. Program may crash when stack overflows.

75. What is static function in C?

All functions are global by default in a C program/file. But, static keyword makes a function as a local function which can be accessed only by the program/file where it is declared and defined. Other programs/files can't access these static functions.

76. If you want to execute C program even after main function is terminated, which function can be used?

- “atexit()” function can be used if we want to execute any function after program is terminated normally.
- Syntax: `int atexit (void (*func)(void));`
- The function “func” will be called automatically without any arguments once program is terminated normally.

77. What is exit() function in C?

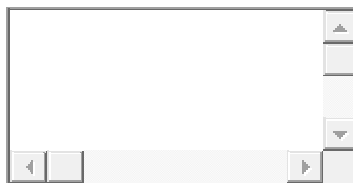
- `exit()` function terminates the program/process normally and returns the status code to the parent program/process.
- Syntax: `void exit(int status)`

78. Is it possible to call atexit() function more than once in a C program?

Yes. We can call `atexit()` function more than once. But, they will be executed in reverse order as a stack.

Example program:

C



```

1 #include <stdio.h>
2 #include <stdlib.h>
3
4 void Exit1 (void)
5 {
6     printf ("Exit1 function is called\n");
7 }
8 void Exit2 (void)
9 {
10    printf ("Exit2 function is called \n");
11 }
12
13 int main (void)
14 {
15     atexit (Exit1);
16     atexit (Exit2);
17     printf ("This is the end of this program.\n");
18     return 0;
19 }

```

Output:

```

This is the end of this program.
Exit2 function is called
Exit1 function is called

```

79. What is the difference between exit() and return() in C?

- `exit()` is a system call which terminates current process. `exit()` is not an instruction of C language.
- Whereas, `return()` is a C language instruction/statement and it returns from the current function (i.e. provides exit status to calling function and provides control back to the calling function).

80. What is the use of “#define” in C?

`#define` is a pre-processor directive which is used to define constant value. This constant can be any of the basic data types.

81. What is the syntax for comments in C?

Below is the syntax for comments in C. The characters or words or anything which are given between “/*” and “*/”, won’t be considered by C compiler for compilation process. These will be ignored by C compiler during compilation.

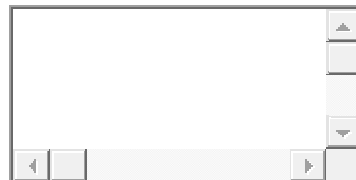
Syntax: /* your comments here */

82. What is “##” operator in C?

`##` is a pre-processor macro in C. It is used to concatenate 2 tokens into one token.

Example program:

C



```
1 #include<stdio.h>
2
3 #define concatenation(a,b) a ## b
4
5 int main ()
6 {
7     int ab = 1000;
8     printf("The concatenated value is:%d \n",concatination(a,b));
9     return 0;
10 }
```

Output: The concatenated value is: 1000

83. How will you override an existing macro in C?

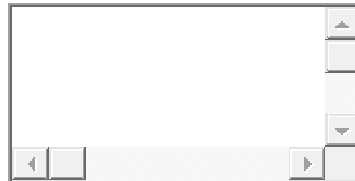
To override an existing macro, we need to undefine existing macro using “`#undef`”. Then, we need to define same macro again with new value.

84. What is pragma in C? Or how will you execute functions before and after main function in C program?

#pragma is a pre-processor macro in C. It is used to call a function before and after main function in a C program.

Example program:

C



```
1 #include <stdio.h>
2
3 void function1( );
4 void function2( );
5
6 #pragma startup function1
7 #pragma exit function2
8
9 int main( )
10 {
11     printf ( "\n Now we are in main function" );
12     return 0;
13 }
14 void function1( )
15 {
16     printf("\nFunction1 is called before main function call");
17 }
18 void function2( )
19 {
20     printf ( "\nFunction2 is called just before end of " \
21             "main function" );
22 }
```

Output:

Function1 is called before main function call
Now we are in main function
Function2 is called just before end of main function

85. How to check whether macro is defined or not in a C program?

"#ifdef" pre-processor directive checks whether particular macro is defined or not in a C program.

86. What is the difference between memcpy() & strcpy() functions in C?

- memcpy() function is used to copy a specified number of bytes from one memory to another. Whereas, strcpy() function is used to copy the contents of one string into another string.
- memcpy() function acts on memory rather than value. Whereas, strcpy() function acts on value rather than memory.

87. What is the difference between memcpy() & memmove() functions in C?

- memcpy() function is used to copy a specified number of bytes from one memory to another.
- memmove() function is used to copy a specified number of bytes from one memory to another or to overlap on same memory.
- Difference between memmove() and memcpy() is, overlap can happen on memmove(). Whereas, memory overlap won't happen in memcpy() and it should be done in non-destructive way.

88. Is there any inbuilt library function in C to remove leading and trailing spaces from a string? How will you remove them in C?

- There is no inbuilt function to remove leading and trailing spaces from a string in C. We need to write our own function to remove them.
- We need to check first non-space character in given string. Then, we can copy that string from where non space character is found. Then, we can check whether any spaces are available in copied string from the end of the string. If space is found, we can copy '\0' in that space until any character is found. Because, '\0' indicates the end of the string. Now, we have removed leading and trailing spaces in a given string.

89. What is the difference between strcpy() & strncpy() functions in C?

- strcpy() function copies whole content of one string into another string. Whereas, strncpy() function copies portion of contents of one string into another string.
- If destination string length is less than source string, entire/specified source string value won't be copied into destination string in both cases.

90. Can array subscripts have negative value in C?

No. Array subscripts should not have negative value. Always, it should be positive.

91. What is the difference between array and string in C?

- Array can hold any of the data types. But, string can hold only char data type.
- Array size must be a constant value. So, array size can't be changed once declared. But, string size can be modified using char pointer.
- Array is not ended with null character by default. But, string is ended with null ('\0') character by default.

92. What is pointer in C?

Pointer is a variable that stores/points the address of another variable. Normal variable stores the value of the variable whereas pointer variable stores the address of the variable.

Syntax: data_type *var_name;

Where, * is used to denote that "p" is pointer variable and not a normal variable.

93. What is null pointer in C?

Null pointer is a pointer which is pointing to nothing. Null pointer points to empty location in memory. Value of null pointer is 0. We can make a pointer to point to null as below.

```
int *p = NULL;
char *p = NULL;
```

94. What is NULL in C?

NULL is a macro which is defined in C header files. The value of NULL macro is 0. It is defined in C header files as below.

```
#define NULL (void *) 0;
```

NULL is used for pointers only as it is defined as (void *) 0. It should not be used other than pointers. If NULL is assigned to a pointer, then pointer is pointing to nothing.

95. What is void pointer in C?

- Void pointer is a generic pointer that can be used to point another variable of any data type.
- Void pointer can store the address of variable belonging to any of the data type. So, void pointer is also called as general purpose pointer.

Note:

int pointer can be used to point a variable of int data type and char pointer can be used to point a variable of char data type.

96. What is dangling pointer in C?

When a pointer is pointing to non-existing memory location is called dangling pointer.

97. What is wild pointer in C?

Uninitialized pointers are called as wild pointers in C which points to arbitrary (random) memory location. This wild pointer may lead a program to behave wrongly or to crash.

98. What is file pointer in C?

- File pointer is a pointer which is used to handle and keep track on the files being accessed. A new data type called "FILE" is used to declare file pointer. This data type is defined in stdio.h file. File pointer is declared as FILE *fp. Where, 'fp' is a file pointer.
- fopen() function is used to open a file that returns a FILE pointer. Once file is opened, file pointer can be used to perform I/O operations on the file. fclose() function is used to close the file.

99. When can void pointer and null pointer be used in C?

Void pointer is a generic pointer that can be used to point another variable of any data type. Null pointer is a pointer which is pointing to nothing.

100. What is const pointer in C?

Const pointer is a pointer that can't change the address of the variable that is pointing to. Once const pointer is made to point one variable, we can't change this pointer to point to any other variable. This pointer is called const pointer.

101. Is pointer arithmetic a valid one? Which arithmetic operation is not valid in pointer? Why?

Pointer arithmetic is not valid one. Pointer addition, multiplication and division are not allowed as these are not making any sense in pointer arithmetic.

But, two pointers can be subtracted to know how many elements are available between these two pointers.

102. Is void pointer arithmetic a valid one? Why?

Arithmetic operation on void pointer is not valid one. Void pointer is a generic pointer. It is not referring int, char or any other data type specifically. So, we need to cast void pointer to specific type before applying arithmetic operations.

Note:

- Pointer arithmetic itself is not valid one. Pointer addition, multiplication and division are not allowed as these are not making any sense in pointer arithmetic.
- But, two pointers can be subtracted to know how many elements are available between these two pointers.

103. What is the difference between null and zero?

NULL is a macro which is defined in C header files. The value of NULL macro is 0. It is defined in C header files as below.

```
#define NULL (void *) 0;
```

- NULL is used for pointers only as it is defined as (void *) 0. It should not be used other than pointers. If NULL is assigned to a pointer, then pointer is pointing to nothing.
- 0 (zero) is a value.

104. What is the difference between null pointer and uninitialized pointer in C?

Null pointer is a pointer which is pointing to nothing. Null pointer points to empty location in memory. Value of null pointer is 0. We can make a pointer to point to null as below.

```
int *p = NULL;  
char *p = NULL;
```

Uninitialized pointers are called as wild pointers in C which points to arbitrary (random) memory location. This wild pointer may lead a program to behave wrongly or to crash.

105. Can array size be declared at run time?

Array size can't be declared at run time. Size of array must be known during compilation time. So, array size should be declared before compilation.

- Correct example: `char array [10];`
- Wrong example: `char array[i];`
- Always, Array subscript should be positive and it should not be either negative or any variable name.
- If we really don't know the size of an array, we can use dynamic memory allocation concepts in C which uses `malloc()`, `calloc()` functions to allocate memory during execution time.

106. What is memory leak in C?

- Memory leak occurs when memory is allocated but not released back to the operating system. Memory leakage increases unwanted memory usage. So, it reduces performance of the computer by reducing available memory.
- In modern operating systems, memory is released back to the operating system when application terminates.

107. What happens when we try to access null pointer in C?

NULL pointer is pointer that is pointing to nothing (No memory location). Accessing null pointer in C may lead a program to crash. So, Null pointer should not be accessed in a program.

108. What is meant by segmentation fault or memory fault in C?

Segmentation fault is a fault that occurs because of illegal/invalid memory access. Illegal memory access means, When a program tries to access a memory location that is not allowed or when a program tries to access a memory location in a way that is not allowed.

109. What is meant by core dump in C?

- Core dump or core is a file, generated when a program is crashed or terminated abnormally because of segmentation fault or some other reason. Information of the memory used by a process is dumped in a file called core. This file is used for debugging purpose. Core dump has file name like "core.<process_id>"
- Core dump file is created in current working directory when a process terminates abnormally. Core dump is a typical error occurs because of illegal memory access.
- Core dump is also called as memory dump, storage dump or dump.

110. Can a pointer be freed more than once in C? What happens if do so? Or can a pointer be freed twice in C?

1st scenario: After freeing a pointer in a C program, freed memory might be reallocated by some other or same program luckily. In this scenario, freeing the same pointer twice won't cause any issue.

2nd scenario: We can free a pointer. Then, we can allocate memory for same pointer variable. Then, we can use it and free it again. This is also not an issue.

3rd scenario: If we free the same pointer second time without reallocating memory to that pointer, then what happens? As per ANSI/ISO C standard, this is undefined behaviour. This undefined behaviour may cause anything to the program that we do not expect to happen.

111. What is the size of int pointer and char pointer in C?

- Pointer variable size is not depending on data type as pointer always stores the address of other variable which is always integer data type.
- So, any pointer (int, char, double, etc) size will be 2 for 16 bit processor, 4 for 32 bit processor and 8 for 64 bit processor.
- sizeof() operator can be used to evaluate size of a variable/pointer in C.

112. How will you print the value and address of a pointer variable (example int *p) in C?

- We can use printf ("%x", p); statement to print the address that pointer "p" stores.
- We can use printf ("%d", *p); statement to print the value of the pointer variable.

113. How will you print the value and address of a normal variable (example int p) in C?

We can use printf ("%x", &p); statement to print the address that pointer "p" stores.

We can use printf ("%d", p); statement to print the value of the normal variable.

114. What are library functions and their use in C language? Can we write our own functions and include them in C library?

- Library functions in C language are inbuilt functions which are grouped together and placed in a common place called library. The use of library function is to get the pre-defined output instead of writing our own code to get those outputs.
- Yes. We can write our own functions and include them in C library.

115. Can variable name be start with underscore in C?

Yes. A variable name can start with underscore in C programming language.

116. Can variable name have special symbols in C?

No. Except underscore (_), other special symbols are not allowed to use for naming the variables in C.

117. What is inline function in C?

- A normal function becomes inline function when function prototype of the function is prepended with keyword "inline".
- The property of inline function is, compiler inserts the entire body of the function in the place where inline function name is used in the program.
- Advantage of inline function is, it does not require function call and does not return anything from the function.
- Disadvantage of inline function is, it increases file size as same function code is copied again and again in the program wherever it is called.

118. Is it possible to print "Hello World" without semicolon in C? How?

Yes. It is possible to print "Hello World" without semicolon in C program. Please refer below example program.

This program prints "Hello World" in console window and returns 11 which is the value of string length.

```
#include <stdio.h>
int main()
{
if(printf("Hello World"))
{
/* Do nothing */
}
}
```

119. What is the scope of local, global and environment variables in C?

- The scope of local variables will be within the function only. These variables are declared within the function and can't be accessed outside the function.
- The scope of global variables will be throughout the program. These variables can be accessed from anywhere in the program. This variable is defined outside the main function. So, this variable is visible to main function and all other sub functions.
- Environment variable is a variable that will be available for all C applications and C programs. We can access these variables from anywhere in a C program without declaring and initializing in an application or C program.

120. Why are we using pointers in C?

- C Pointer is a variable that stores/points the address of another variable.
- C Pointer is used to allocate memory dynamically (i.e. at run time)

121. What is static variable in C?

Static variable is a variable that retains its value between different function calls.

122. What is extern and static function in C?

- By default, any function that is defined in a C file is extern. These functions can be used in any other source file of the same project which has many other files.
- When we declare/define a function as static, these functions can't be used in other files of the same project.
- Also, if we want to use the same function name in different files of the same project, we can use static function which won't through any compilation error for duplicate function name.

123. What is the difference between pointer and array in C?

- Array is a collection of variables belonging to the same data type. We can store group of data of same data type in an array.
- Pointer is a single variable that stores the address of other object/variable.

124. Can a variable be both volatile and constant in C?

Yes. A variable can be declared as both volatile and constant in C.

Const modifier does not allow changing the value of the variable by internal program. But, it does not mean that value of const variable should not be changed by external code. So, a variable can be both volatile and constant in C.

125. What is the use of main() function in C?

main() function is the function from where execution of any C program begins. So, main() function is mandatory for any C program.